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Overcoming limitations on gate fidelity in noisy static exchange-coupled surface qubits

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Recent experiments demonstrated that the spin state of individual atoms on surfaces can be quantum-coherently controlled through all-electric electron spin resonance. By constructing interacting arrays of atoms, this results in an atomic-scale qubit platform. However, the static exchange coupling between qubits, limited lifetime, and polarization of the initial state impose significant limits on high-fidelity quantum control. We address this issue using open quantum systems simulation and quantum optimal control theory. We demonstrate the conditions under which high-fidelity operations ($\mathcal{F} \gtrsim 0.9$) are feasible in this qubit platform, and show how the Krotov method of quantum optimal control theory adapts to specific noise sources to outperform the conventional Rabi drivings. Finally, we re-examine the experimental setup used in the initial demonstration of this qubit platform and propose optimized experimental designs to maximize gate fidelity in this platform.

Atomic-scale exchange-coupled spin qubits have recently emerged as a platform for quantum-coherent nanoscience and quantum information processing^{1,2}. This is enabled by scanning tunneling microscopy (STM), which offers bottom-up fabrication of tailored spin systems with atomic precision³. When combined with electron spin resonance (ESR) techniques^{4,5}, STM allows for the coherent manipulation and readout of a single electron spins^{6,7}. In ESR-STM, an atomically sharp magnetic tip is positioned over a magnetic atom adsorbed on an insulating layer. By applying pulsed radio-frequency (RF) voltages, transitions between spin states can be induced, facilitating coherent control through Rabi oscillations. This setup enables the implementation of arbitrary single-qubit rotations and thus universal control of the spin qubits inside the tunnel junction⁸. To extend coherent control to multiple spins, additional atoms are placed outside of the tunnel junction, and exchange interactions between neighboring spins are used to facilitate readout of each spin state by frequency addressing^{9–11}. When the applied radio-frequency pulses match the resonant conditions of these transitions, their states can be coherently driven, allowing operation of multi-qubit systems^{1,12}.

Despite the significant advancements in these atomic-scale spin qubits, there are several challenges. A significant one is the limited information about the fidelity of gate operations in previous experiments due to the difficulty of quantum-state tomography in this platform. Another limiting factor is the *always-on* static exchange coupling between spins, which induces a complex time-evolution of the coupled quantum system. Experimentally challenging is also radio-frequency crosstalk, where an RF pulse intended to drive a

specific qubit inadvertently affects neighboring qubits when their separation in frequency space is close to the natural linewidth. This phenomenon, also known as cross-resonance, is prevalent in other platforms such as superconducting qubits¹³, quantum dot spin qubits^{14,15}, and nuclear magnetic resonance¹⁶. The problem becomes more pronounced when multiple pulses are applied simultaneously. All of this is compounded by rapid decoherence, arising from the interaction of qubits with their environment. This interaction leads to relaxation of spin states, characterized by the energy relaxation time T_1 , as well as dephasing, characterized by coherence time T_2 ^{17,18}. In this context, the dynamics of the open quantum system can be described using the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation, which incorporates phenomenological collapse operators to model the dissipative effects of the environment^{10,19}.

Strategies to mitigate some of these challenges have been proposed. For instance, to mitigate RF crosstalk, strategies such as designing qubits with optimized resonant frequency allocation²⁰, cross-resonance architecture with lightweight resonator coupler²¹, and synchronizing Rabi oscillations have been employed²². However, these methods do not universally resolve the issues of decoherence. There also exist strategies to cope with decoherence, such as decoherence-free subspaces^{23,24}, noiseless subsystems²⁵, dynamical decoupling²⁶, and bath-optimal minimal-energy control²⁷. Yet, these methods are often restricted to specific interactions between system and environment. Quantum optimal control theory (QOCT)^{28,29} offers a more general approach to address these challenges. QOCT aims to find the optimal control fields that drive the system's evolution toward a desired

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unitary transformation^{30,31}. This approach has been implemented in various systems such as nuclear magnetic resonance^{32,33} and molecular spin qubits³⁴.

In this work, we are going to employ the Krotov method^{35–37} from QOCT for realizing high-fidelity quantum gate operations in static exchange-coupled surface qubits. Our results show that, within certain parameter regimes, high-fidelity control ($\mathcal{F} \gtrsim 0.9$) is feasible even in the presence of energy relaxation and decoherence. In addition, optimized pulses obtained via the Krotov method result in significantly improved fidelity by exploiting all degrees of freedom of the control Hamiltonian, and it delivers noise-adapted pulses that under noisy conditions outperform those optimized without noise. Using parameters close to recent experiments, we demonstrate that average fidelities of 0.989 for the NOT gate and 0.887 for the CNOT gate can be achieved in two- and three-spin qubit systems, respectively. Besides gate operations, we apply the method to entanglement preparation and show that the fidelity approaches the upper bound set by the concurrence, and it is largely limited by the thermal population of the initial state. Finally, we revisit the experimental setup used in the first demonstration of this static exchange-coupled surface qubits platform and propose improvements guided by our simulations. Our work demonstrates a promising usage of QOCT in this miniature qubit platform.

Results

Static exchange-coupled on-surface qubits

We first consider a generic model for an on-surface qubit system. The drift Hamiltonian describes N spin centers (each with electron spin $S = 1/2$) in external magnetic fields and connected via Heisenberg-like exchange coupling. The control Hamiltonian contains the time-dependent magnetic fields at relevant radio-frequencies. The total Hamiltonian reads

$$H(t) = - \sum_{i=1}^N \omega_i S_i^z + \sum_{i=1}^N \sum_{j>i}^N J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_i \sum_k^{N_k} \Omega_k \cos(\omega_{\text{RF}k} \tau_k + \phi_k) \sigma_i^x, \quad (1)$$

where the first term describes Zeeman energy with Larmor frequencies ω_i set by local magnetic fields $\mathbf{B}_i$ (taking the z -axis as quantization axis defined by the local magnetic field such that the Larmor term is proportional to S_i^z , and $\omega_i = g\mu_B B_i$ with μ_B Bohr magneton), the second term is the Heisenberg isotropic exchange between qubits, which in on-surface qubits is always on and static, i.e., it cannot be controllably switched on and off as in other platforms^{38,39}. The final term describes the driving pulses using time-varying perpendicular magnetic fields. $\mathbf{S}_i = \{S_i^x, S_i^y, S_i^z\} = \hbar/2\{\sigma_i^x, \sigma_i^y, \sigma_i^z\}$ is the spin operator vector of the i -th qubit, with S_i^α ($\alpha = x, y, z$) as the Pauli spin matrices. N_k denotes the number of simultaneous pulses applied to the system, with $\omega_{\text{RF}k}$, Ω_k , τ_k , being the frequency, amplitude, and length of the k -th Rabi driving pulse, respectively. Note that we set the phase $\phi_k = 0 \forall k$ throughout for simplicity, but phase differences between driving pulses can, in principle, affect the gate fidelity (see Supplementary Material Fig. S1).

Limits on gate fidelity in the absence of decoherence

The presence of coupling terms inherently limits the fidelity of quantum gates even in the decoherence-free limit. We consider a closed quantum system and discuss the effects of relaxation and decoherence later. To show this, we examine a two-qubit system, as illustrated in Fig. 1a, for a realization of the NOT gate on the second qubit. The system's Hamiltonian in Eq. (1) can be simplified as

$$H_2(t) = -\omega_1 S_1^z - \omega_2 S_2^z + J \mathbf{S}_1 \cdot \mathbf{S}_2 + [\Omega_1 \cos(\omega_{\text{RF}1} t) + \Omega_2 \cos(\omega_{\text{RF}2} t)](\sigma_1^x + \sigma_2^x). \quad (2)$$

We first need to solve the eigenstates of the drift Hamiltonian, i.e., the time-independent part of Eq. (2). The Heisenberg coupling term hybridizes $|01\rangle$ and $|10\rangle$ states and leaves the other two, $|00\rangle$ and $|11\rangle$, unchanged. Thus, the basis for the drift Hamiltonian is $\{|00\rangle, |\tilde{0}1\rangle, |\tilde{1}0\rangle, |11\rangle\}$, with

$$\begin{aligned} |\tilde{0}1\rangle &= -\sin \xi_1 |01\rangle + \cos \xi_1 |10\rangle, \\ |\tilde{1}0\rangle &= \sin \xi_2 |01\rangle + \cos \xi_2 |10\rangle, \end{aligned} \quad (3)$$

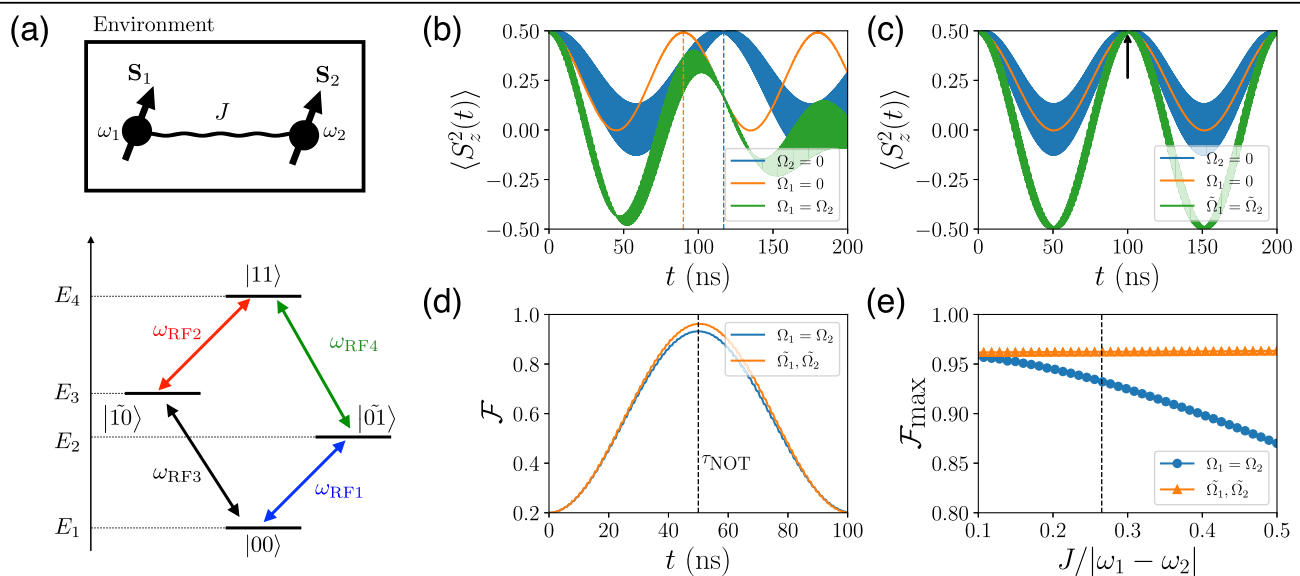


Fig. 1 | Illustration for the realization of the NOT gate in the closed system.

a Schematic of the closed system of two on-surface qubits, i.e., their interaction with the surrounding environment is hypothetically ruled out. J denotes their coupling strength, and $\mathbf{S}_i = \{S_i^x, S_i^y, S_i^z\}$ is the spin operator vector of the i th qubit. Different Larmor frequencies are realized by different local magnetic fields, $\omega_{1,2} = g\mu_B B_{1,2}$. The energy diagram and frequencies corresponding to a single-qubit flip are plotted. Analytic forms of eigenenergies E_1 to E_4 are in Eq. (5) of the main text. $\omega_{\text{RF}i}$ indicates the transition frequency of the i -th transition. **b** Spin evolution of the second qubit in the lab frame is plotted. The initial state is $1/\sqrt{2}(|00\rangle + |\tilde{1}0\rangle)$. The blue and orange lines are with respect to transitions

$|00\rangle \leftrightarrow |\tilde{0}1\rangle$ ($\Omega_2 = 0$) and $|\tilde{1}0\rangle \leftrightarrow |11\rangle$ ($\Omega_1 = 0$). The parameters of the drift Hamiltonian are $\omega_1 = 20\pi$ GHz, $\omega_2 = 14\pi$ GHz, $J = 5$ GHz. Driving amplitudes are $\Omega_1 = \Omega_2 = \pi/50$ GHz. Vertical dashed lines mark the peak positions of the corresponding transitions. **c** The spin evolution under Rabi rate synchronization shown in Eq. (9). The black arrow indicates the alignment of the two peaks. **d** Fidelity of NOT gate on the second qubit as a function of time. τ_{NOT} indicates the expected time to realize the NOT gate with given driving amplitudes Ω_1 and Ω_2 . **e** Fidelity of the NOT gate with and without using Rabi rate synchronization as a function of coupling strength J . Vertical dashed line indicates the coupling strength used in previous figures.

where

$$\begin{aligned}\xi_1 &= \arctan(\alpha + \sqrt{\alpha^2 + 1}), \\ \xi_2 &= \arctan(-\alpha + \sqrt{\alpha^2 + 1}),\end{aligned}\quad (4)$$

and $\alpha \equiv (\omega_1 - \omega_2)/J$ is the dimensionless parameter quantifying the strength of coupling with respect to Larmor frequencies. The eigenenergies in increasing order are

$$\begin{aligned}E_1 &= \frac{J}{4} - \frac{\omega_1 + \omega_2}{2}, \\ E_2 &= \frac{\omega_1 - \omega_2}{2} \cos 2\xi_1 - \frac{J}{4}(1 + 2 \sin 2\xi_1), \\ E_3 &= \frac{\omega_1 - \omega_2}{2} \cos 2\xi_2 + \frac{J}{4}(-1 + 2 \sin 2\xi_2), \\ E_4 &= \frac{J}{4} + \frac{\omega_1 + \omega_2}{2}.\end{aligned}\quad (5)$$

The NOT gate on the second qubit requires transitions $|00\rangle \leftrightarrow |\tilde{0}1\rangle$ and $|\tilde{1}0\rangle \leftrightarrow |11\rangle$, as illustrated in Fig. 1a. Thus, the corresponding transition frequencies are $\omega_{\text{RF1}} = E_2 - E_1$ and $\omega_{\text{RF2}} = E_4 - E_3$, and the driving pulse can be presented as a bichromatic pulse

$$\mathcal{D}(t) \equiv \Omega_1 \cos(\omega_{\text{RF1}} t) + \Omega_2 \cos(\omega_{\text{RF2}} t), \quad (6)$$

where $\Omega_{1(2)}$ are the amplitudes. We note that a controlled NOT gate requires a monochromatic pulse only, that is, ω_{RF2} for the second qubit as the target one¹.

The matrix representation of the Hamiltonian in Eq. (2) in the eigenstate basis can be written as follows

$$\mathbf{H}_2(t) = \begin{matrix} & |00\rangle & |\tilde{0}1\rangle & |\tilde{1}0\rangle & |11\rangle \\ \begin{matrix} |00\rangle \\ |\tilde{0}1\rangle \\ |\tilde{1}0\rangle \\ |11\rangle \end{matrix} & \begin{bmatrix} E_1 & (\cos \xi_1 - \sin \xi_1)\mathcal{D}(t) & (\cos \xi_2 + \sin \xi_2)\mathcal{D}(t) & 0 \\ (\cos \xi_1 - \sin \xi_1)\mathcal{D}(t) & E_2 & 0 & (\cos \xi_1 - \sin \xi_1)\mathcal{D}(t) \\ (\cos \xi_2 + \sin \xi_2)\mathcal{D}(t) & 0 & E_3 & (\cos \xi_2 + \sin \xi_2)\mathcal{D}(t) \\ 0 & (\cos \xi_1 - \sin \xi_1)\mathcal{D}(t) & (\cos \xi_2 + \sin \xi_2)\mathcal{D}(t) & E_4 \end{bmatrix} \end{matrix} \quad (7)$$

There are two potential issues that can limit the fidelity of the NOT gate, as discussed below. The first is the crosstalk between driving pulse frequencies. If the two resonant frequencies are close, $\omega_{\text{RF1}} \approx \omega_{\text{RF2}}$, each pulse can off-resonantly drive the other's transition, resulting in a low fidelity NOT gate. A solution for this issue is to ensure that the coupling strength J is sufficiently large so that the resonant frequencies are well separated.

The second challenge is the amplitude mismatch between the two transitions involved in the NOT gate. The NOT gate is realized by setting equal Rabi driving amplitude, $\Omega_1 = \Omega_2$, with a pulse width $\tau_{\text{NOT}} = \pi/\Omega_1$ for the remote qubit being flipped. However, as implied in Eq. (7), the transitions have different effective driving amplitudes induced by the Heisenberg coupling, respectively characterized by

$$\begin{aligned}\tilde{\Omega}_1 &= \Omega_1(\cos \xi_1 - \sin \xi_1), \\ \tilde{\Omega}_2 &= \Omega_2(\cos \xi_2 + \sin \xi_2).\end{aligned}\quad (8)$$

The mismatch results in unequal periods of transitions depicted in Fig. 1b and thus worsens the fidelity of NOT gates. A solution for this issue is *Rabi rate synchronization*⁴⁰, where the Rabi driving amplitudes are adjusted so that the effective amplitude equal to each other $\tilde{\Omega}_1 = \tilde{\Omega}_2$. To achieve Rabi rate synchronization, the two amplitudes should obey the following condition:

$$\frac{\Omega_1}{\Omega_2} = \frac{\cos \xi_2 + \sin \xi_2}{\cos \xi_1 - \sin \xi_1}. \quad (9)$$

Using the drift Hamiltonian as specified in the caption of Fig. 1 results in a ratio of $\Omega_1/\Omega_2 \approx 1.3$. The two hybridized eigenstates are close to Zeeman basis states, $|\tilde{0}1\rangle \approx 0.992|01\rangle - 0.129|10\rangle$ and $|\tilde{1}0\rangle \approx 0.129|01\rangle + 0.992|10\rangle$.

The transitions with synchronized Rabi rates are visualized in Fig. 1c, and how it improves the fidelity of NOT gates is shown in Fig. 1d. Note that the above ratio of synchronized amplitude is fully determined via coupling strength with respect to Larmor frequencies.

We extend the analysis with different values of coupling strength J . The results are presented in Fig. 1e. As we increase J , the mismatch is more severe and leads to lower fidelity. Rabi rate synchronization helps to correct the mismatch and yield nearly constant fidelity over coupling strength. This is because Rabi rate synchronization only addresses the mismatch issue, while there exist other unwanted transitions, $|00\rangle \leftrightarrow |\tilde{1}0\rangle$ and $|\tilde{0}1\rangle \leftrightarrow |11\rangle$, as seen in Eq. (7). Even though the frequencies of these transitions are off-resonant, they are still driven by the pulse, albeit with small coefficients. This results in leakage of population to unwanted states and thus limits the fidelity of NOT gates.

We have seen that even in the simplest case of a two-qubit system, it is not possible to fully mitigate all the issues raised by exchange coupling, and there exists an upper bound on the achievable fidelity. The complexity increases exponentially as more qubits become involved in the dynamics: it becomes difficult to design well-separated peaks and to solve the Rabi rate synchronization.

QOCT in the decoherence-free limit

Having identified coherent fidelity limits of monochromatic/bichromatic Rabi driving caused by always-on exchange coupling (crosstalk,

amplitude renormalization, and leakage), we now ask whether these errors can be removed by optimizing the full control waveform. We first answer this in the decoherence-free (closed-system) limit, where any remaining infidelity is purely coherent, before moving to optimization in the presence of noise. Using the same drift Hamiltonian as above, we apply the Krotov method (see “Methods” section for details) to realize a NOT gate on the second qubit at fixed gate time and with near-unit fidelity.

To set an identical condition for comparison, we set the width for the optimal pulse to be $\tau = 50$ ns, which is the same as the pulse width for the Rabi rate synchronization method. We use a flat-top pulse as the initial guess for optimization. The Krotov method parameters are $\lambda_a = 10$ and $N_{\text{iter}} = 600$. The optimal pulse obtained after 600 iterations is presented in Fig. 2a. It is more relevant to analyze the frequency domain obtained via the Fourier transform of the optimal pulse, as shown in Fig. 2b. The spectrum reveals four pronounced peaks located at $\omega_{\text{RF1},2,3,4}$, which are resonant frequencies corresponding to single-qubit flips of the drift Hamiltonian (see Fig. 1a). This is in contrast to the Rabi rate synchronization method, where the spectrum obviously has only two peaks at ω_{RF1} and ω_{RF2} . One can consider that the initial flat-top pulse comprises all frequencies, and the optimization process with the Krotov method selects the relevant ones. The optimal pulse found via the Krotov method shows a realization of the NOT gate with unit fidelity (up to numerical error), as shown in Fig. 2c. In addition, the inset indicates that the optimal pulse can achieve nearly perfect fidelity for any coupling strength J as long as the Larmor frequency difference is non-zero and J is large enough to clearly separate the individual transition energies (see also Supplementary Material, Fig. S5). This result suggests that the

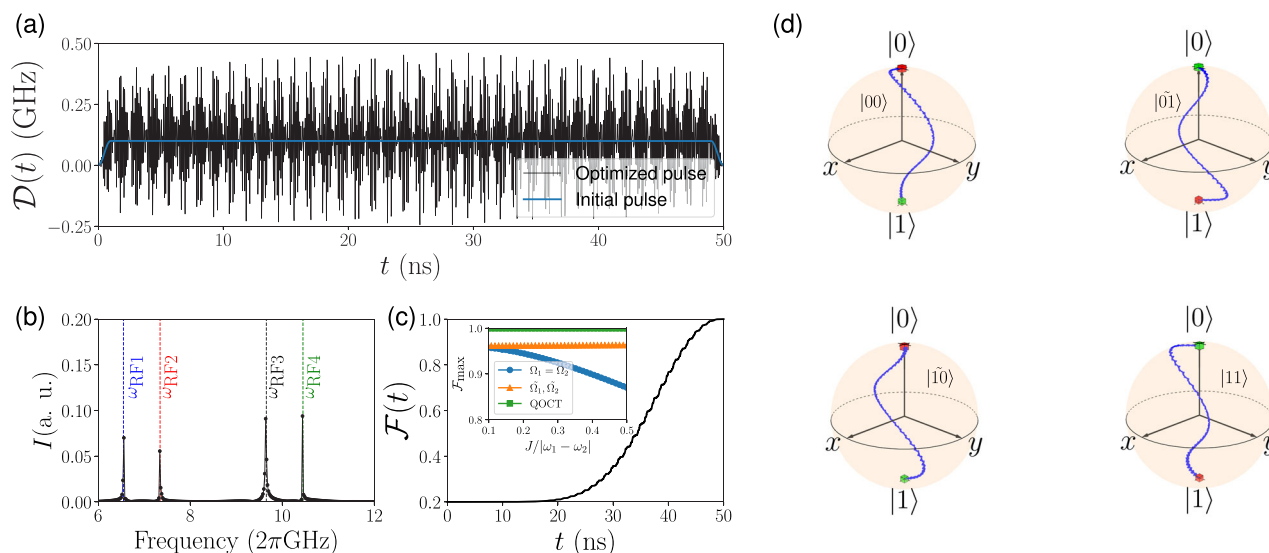


Fig. 2 | Realization of a NOT gate on the second qubit using QOCT. **a** The optimal pulse (black) compared to the initial flat pulse (blue). The pulse width is set to $\tau = 50$ ns. The drift Hamiltonian parameters are $\omega_1 = 20\pi$ GHz, $\omega_2 = 14\pi$ GHz, and $J = 5$ GHz. The Krotov method parameters are $\lambda_a = 10$ and $N_{\text{iter}} = 600$. **b** Fourier transform of the optimal pulse. **c** Fidelity of the NOT gate as a function of time, reaching nearly 1 at the end

of the optimal pulse. The inset indicates that the unit fidelity can be achieved using Krotov with all values of coupling J . **d** Evolution of the second qubit's state on the Bloch sphere in the interaction picture for all basis states. Red and green cubes represent the initial and final states, respectively.

Krotov method utilizes all degrees of freedom in the control Hamiltonian to achieve high-fidelity quantum gates.

We further illustrate the spin evolution of the second qubit, corresponding to four initial states, on the Bloch sphere in Fig. 2d. The spin trajectory is computed from the reduced density matrix, with fast oscillations removed using the interaction picture (see “Methods” section for details). It is worth noting that the vector of the final state is slightly smaller than unit length due to the partial mixing of eigenstates.

To support our argument on the utilization of all degrees of freedom, we have also analyzed two different cases. One case is the realization of a CNOT gate, which, in principle, requires only $\omega_{\text{RF}2}$. The other case is with the Ising interaction, where there are no transverse interactions to alter the driving amplitude felt by the system. In other words, a situation in which the NOT gate can be realized with unit fidelity even by implementing a bichromatic pulse $\omega_{\text{RF}1}$ and $\omega_{\text{RF}2}$ with equal amplitude. In both cases, the optimal pulse is still able to achieve nearly perfect fidelity, and its spectrum still shows four pronounced peaks at the resonant frequencies. The details of these two cases are provided in the Supplementary Material (Fig. S4). In addition, we have also employed a gradient-based method of QOCT, GRAdient Ascent Pulse Engineering (GRAPE)³², to optimize the NOT gate in closed systems (Fig. S6). In closed systems, both GRAPE and Krotov achieve near unity fidelity. We found that the spectrum of the optimal pulse obtained via GRAPE is strongly dependent on the initial guess pulse and often uses frequencies that don't correspond to the natural resonances of the system, making an analysis in the context of this work difficult (Figs. S7 and S8). The details are discussed in the Supplementary Material.

QOCT in the presence of decoherence

The results for the closed system establish that, once the full waveform is optimized, the coherent limitations imposed by always-on exchange coupling can be largely removed, enabled by a control spectrum that resembles the relevant system resonances. In experiments, however, gate performance is ultimately limited not by coherent leakage but by irreversible coupling to the environment during the control pulse. To estimate the limitations imposed by the environment, we extend the Krotov optimization to open-system dynamics and quantify how energy relaxation and pure dephasing reshape both the achievable fidelities and the structure of the optimal control fields. We model the dynamics with a GKSL master equation and consider

the two dominant channels, energy relaxation and pure dephasing, separately to isolate their individual impact on the achievable fidelity.

The relaxation and pure dephasing processes are characterized by relaxation time T_1 and pure dephasing time T_ϕ , respectively. The overall decoherence rate, which is the inverse of coherence time T_2 , for a particular k -th spin can be expressed as

$$\frac{1}{T_2^{(k)}} = \frac{1}{2T_1^{(k)}} + \frac{1}{T_\phi^{(k)}}. \quad (10)$$

We are going to investigate the two decoherence mechanisms independently. For pure dephasing, we set $T_1^{(k)} \rightarrow \infty$, while for relaxation, we set $T_\phi^{(k)} \rightarrow \infty$. The drift Hamiltonian parameters are set to be identical as in the previous section.

In addition, it is worth noting that in the ESR-STM platform, typically, the qubits have *inherently asymmetric energy relaxation rates*, as illustrated in Fig. 3a. In such a system, the sensor spin is subject to the tunneling electrons, which is necessary to facilitate the readout but inevitably leads to $T_2^{(1)} \ll T_2^{(2)}$.

Figure 3b and c show the optimized fidelity of a NOT gate on a remote qubit with respect to relaxation and dephasing mechanisms, respectively. A representative value of pulse width $\tau = 10$ ns is used. The optimal fidelity in both mechanisms shares expected features that fidelity increases as the coherence time increases. However, the properties of optimized pulses show subtle differences between these two cases, which will be discussed in the following section. In the limit of infinitely long coherence time, we recover unit fidelity as in closed systems. As the asymmetry in coherence time is imposed, the optimal fidelity is bounded by the coherence time of the sensor qubit, i.e., $T_1^{(1)}$ or $T_\phi^{(1)}$. The upper bound of optimal fidelity depends on the relative coherence time of the sensor qubit with respect to the pulse width. For instance, if $T_1^{(1)} \gg \tau$ or $T_\phi^{(1)} \gg \tau$, the optimal fidelity can be nearly 100%. However, if $T_1^{(1)}$ is comparable to τ , the optimal fidelity will be significantly decreased.

QOCT at finite temperatures: impact on entanglement preparation

The discussion so far focused on gate implementation, where decoherence during the pulse limits the achievable average gate fidelity. A

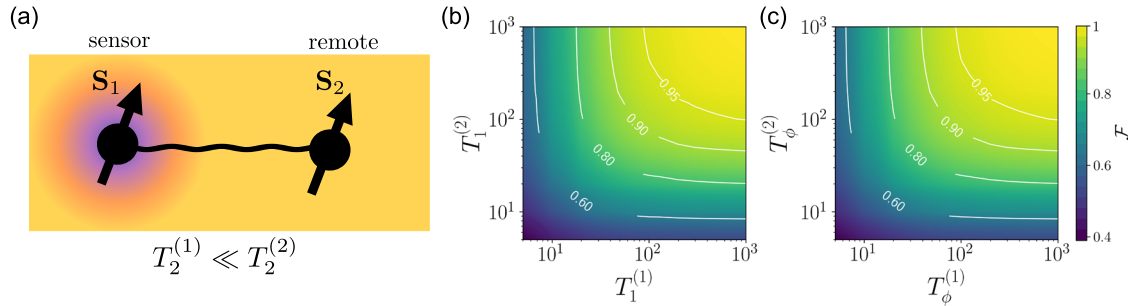


Fig. 3 | Optimized fidelity of a NOT gate on a remote qubit with energy relaxation and pure dephasing. **a** Illustration for the inherent asymmetry in coherence time of qubits. The sensor qubit subject to the tunnel current has a much shorter lifetime, whilst the remote qubit is only subject to substrate scattering, i.e., $T_2^{(1)} \ll T_2^{(2)}$.

Optimal fidelity of NOT gate on remote qubit obtained with Krotov method for **b** energy relaxation and **c** pure dephasing. Pulse width is $\tau = 10$ ns. The parameters of the drift Hamiltonian are $\omega_1 = 20\pi$ GHz, $\omega_2 = 14\pi$ GHz, $J = 5$ GHz, $T = 0.1$ K.

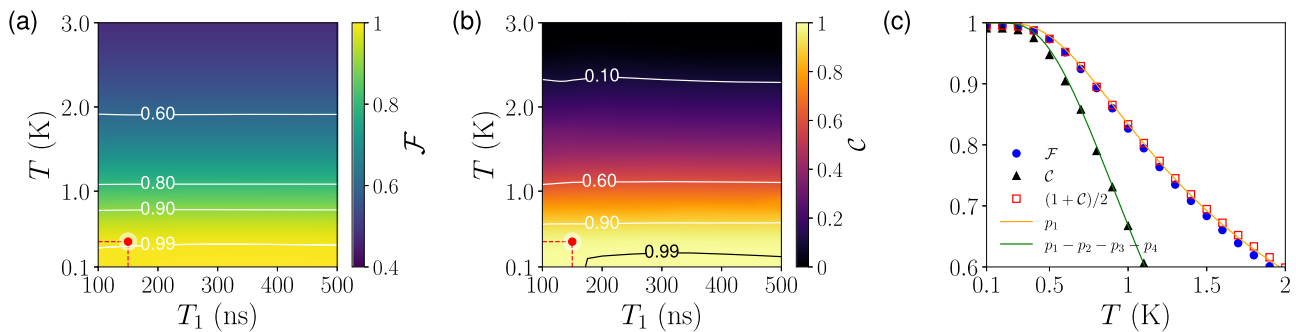


Fig. 4 | Optimized state-to-state preparation for a maximally entangled two-qubit state. **a** Fidelity and **b** Concurrence of the final state as a function of relaxation time T_1 and temperature T , assuming symmetric lifetime $T_1^{(1)} = T_1^{(2)}$. We employ the Krotov method for state-to-state transition from a thermal equilibrium state at temperature T to the Bell state $|\Phi^+\rangle$. **c** Comparison of $\mathcal{F}$ and $\mathcal{C}$ with the Boltzmann factors of the eigenstates of the drift Hamiltonian at different temperatures. Filled

blue circles and filled black triangles indicate the fidelity and concurrence at $T_1 = 200$ ns, respectively. Empty red squares indicate the upper bound of fidelity for a given concurrence (see main text for details). The parameters of the drift Hamiltonian are $\omega_1 = 20\pi$ GHz, $\omega_2 = 14\pi$ GHz, and $J = 5$ GHz. A flattop pulse is used as the initial guess, and the Krotov method parameters are $\lambda_a = 0.01$ and $N_{\text{iter}} = 200$. The pulse width is $\tau = 50$ ns.

complementary task is the preparation of entangled states, where limitations can already arise from the temperature-limited purity of the initial state. To understand the limitations imposed by finite temperature, we simulated state-to-state transfer to create maximally entangled Bell states starting from thermally mixed states⁴¹. We implemented this state-to-state transfer using the same Krotov-based optimization^{35,36} (see “Methods” section). We separately vary T and the relaxation time T_1 to disentangle the roles of thermal population and finite lifetime during the pulse. State-to-state transfer is a natural choice to achieve the optimized pulse, as its cost function is less constrained compared with that of gate optimization^{42,43}.

The initial state is a density matrix at thermal equilibrium of temperature T , of which is given by

$$\rho_0 = \frac{e^{-H_0/k_B T}}{\text{Tr}[e^{-H_0/k_B T}]} = \sum_{i=1}^4 p_i |\psi_i\rangle\langle\psi_i|, \quad (11)$$

where k_B is the Boltzmann constant, $|\psi_i\rangle$ are the eigenstates $\{|00\rangle, |\tilde{0}\tilde{1}\rangle, |\tilde{1}\tilde{0}\rangle, |11\rangle\}$ of the drift Hamiltonian H_0 in Eq. (2), and $p_i = \exp(-E_i/k_B T) / \sum_{j=1}^4 \exp(-E_j/k_B T)$ are the Boltzmann factors with E_i being the corresponding eigenenergies (see Eq. (5)). The target state is one of the four Bell states $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. In addition to the fidelity between the Bell state and the final state (obtained via the Krotov method), we use the concurrence $\mathcal{C}$ of the final state as an additional metric. Concurrence quantifies entanglement of two qubits: concurrence of fully separated states is zero, and concurrence of maximally entangled states is one.

Table 1 | Fidelity $\mathcal{F}$ and concurrence $\mathcal{C}$ for each Bell state and average fidelity at temperature $T = 0.4$ K and $T_1^{(1)} = T_1^{(2)} = 150$ ns

Bell State	$\mathcal{F}$	$\mathcal{C}$
$ \Phi^+\rangle = \frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	0.9849	0.9712
$ \Phi^-\rangle = \frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$	0.9878	0.9760
$ \Psi^+\rangle = \frac{1}{\sqrt{2}}(\tilde{0}\tilde{1}\rangle + \tilde{1}\tilde{0}\rangle)$	0.9816	0.9719
$ \Psi^-\rangle = \frac{1}{\sqrt{2}}(\tilde{0}\tilde{1}\rangle - \tilde{1}\tilde{0}\rangle)$	0.9853	0.9782
Average	0.9849	0.9743

Figure 4a and b present the results of the preparation of Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ as a function of the lifetime of each qubit and temperature, using optimization for state-to-state transfer. We observe that both fidelity and concurrence do not depend on lifetime and are mainly determined by temperature. High fidelity and concurrence are achieved at low temperatures. This result holds for other Bell states, as summarized in Table 1.

The strong dependence of fidelity and concurrence on temperature can be understood by analyzing the Boltzmann factors of the eigenstates of the drift Hamiltonian at different temperatures, as shown in Fig. 4c. At low temperatures, the initial state is predominantly in the ground state $|00\rangle$, where the Krotov method can efficiently drive the system to the target Bell state $|\Phi^+\rangle$. This claim is supported by the fact that fidelity and the

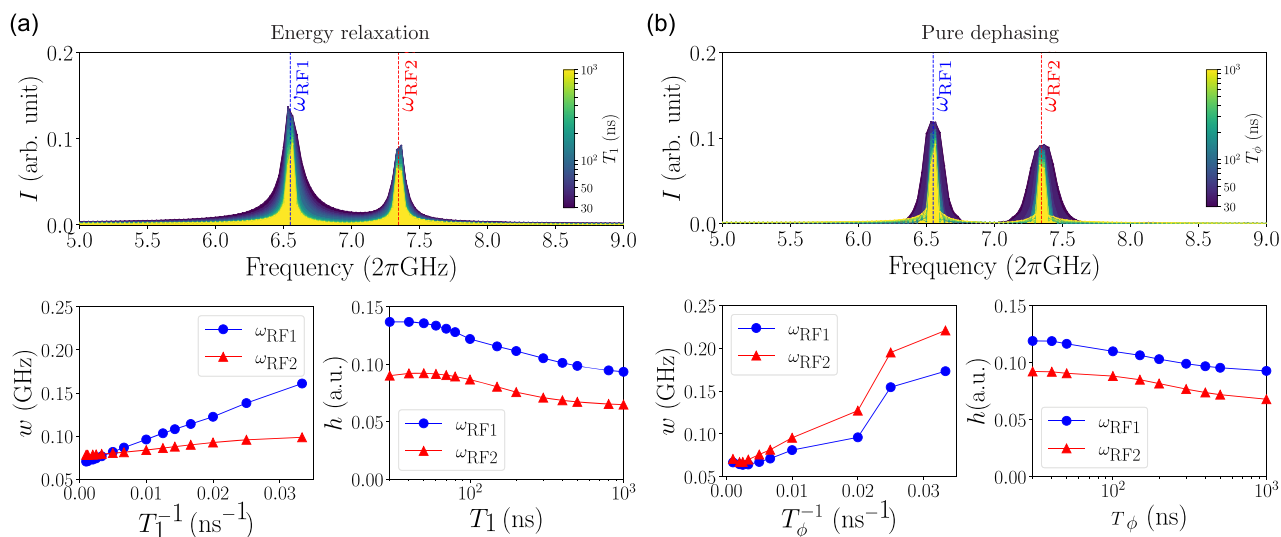


Fig. 5 | Spectrum of the optimized pulses. Properties of the spectrum of the optimized pulse for a NOT gate on a remote qubit with respect to **a** energy relaxation and **b** pure dephasing. The evolution of the spectrum with respect to the corresponding coherence time, and its width and height at ω_{RF1} and ω_{RF2} are shown. The

parameters of the drift Hamiltonian are $\omega_1 = 20\pi$ GHz, $\omega_2 = 14\pi$ GHz, $J = 5$ GHz, $T = 0.1$ K. The pulse width is fixed with $\tau = 30$ ns. A flattop pulse is used as the initial guess, and $N_{\text{iter}} = 100$.

population of ground state p_1 exhibit similar trends as temperature varies. In addition, the fidelity obtained from the Krotov method is close to its upper bound for a given concurrence⁴⁴

$$\mathcal{F} \leq \frac{1 + \mathcal{C}}{2}. \quad (12)$$

The match between concurrence and $p_1 - p_2 - p_3 - p_4$ is a derived result from the above equation, using the normalization condition $\sum_i p_i = 1$.

Discussions

Our results highlight the intrinsic limits of achieving high-fidelity quantum gate operations in static exchange-coupled qubits. In the following, we will discuss (i) how the spectrum of optimized pulses is reshaped by decoherence and (ii) the limits on optimized fidelity with realistic parameters.

A central practical question is how QOCT improves fidelity. This insight should guide future experimental pulse designs. To build this intuition, we analyze the control fields in the frequency domain and track how decoherence reshapes the spectral content of the Krotov-optimized pulses. Figure 5 summarizes how energy relaxation and pure dephasing reshape the spectrum of the Krotov-optimized control. Throughout, we quantify the *width* (w) by the full width at half maximum (FWHM) of each spectral line and the *height* (h) of the corresponding peak amplitude. As in the closed-system case discussed above, the optimal spectrum indicates that the Krotov method exploits the full set of available control degrees of freedom; this behavior persists even with relaxation. We do not consider the long-coherence-time limit here, because the optimization setup differs fundamentally from the closed-system case. Instead of a trotterized exponential unitary propagator, we evolve density matrices using the Lindblad master equation, employing three encoded density matrices to reduce computational cost⁴⁵. While a finite coherence time in the Markovian Lindblad formalism will induce a decrease in fidelity regardless of the shape of the applied pulse⁴⁶, by acting on the pulse-/Hamiltonian-dependence of the average gate fidelity⁴⁷, an optimized pulse can mitigate this decrease, as illustrated in the Supplementary Material Fig. S16.

Across both channels, the optimized pulses remain resonant as there is no discernible peak drift from the transition frequencies, while all the peak envelopes broaden with larger error rates. In particular, for energy relaxation, the amplitude of the peaks appears to remain more or less constant, while for pure dephasing, the peak height increases as dephasing strengthens, as

illustrated in the lower-right panels of Fig. 5a and b. The increase in bandwidth seems to increase monotonically, linearly even, with T_1 (resp. T_ϕ^{-1}) for energy relaxation (resp. pure dephasing), as illustrated in the lower-left panels of Fig. 5a and b. Dephasing broadens the system's susceptibility⁴⁸, one can expect a broader control spectrum to overlap that broadened response better and thus be more robust to phase randomization and energy relaxation. We would like to further note that due to the finite pulse duration, the Fourier transform of the pulse has a minimum width on the order of $1/\tau$, which explains the coarse appearance of the spectrum profile.

Physically, amplitude increase concentrates more energy near resonance, while broadened spectral lobes could indicate, by Fourier duality, faster time-domain features. In the Supplementary Material Fig. S13, a saturation in pulse width is observed for the energy relaxation channel with a Krotov functional weighted differently; this could be explained by the running cost in the Krotov functional (see Eq. (25)), which can act as a soft amplitude cap: once the marginal fidelity gain from making the pulse stronger or more jagged equals the marginal penalty from the running cost, the gradient balances and the update stalls.

These trends suggest clear strategies for improving fidelity at fixed gate duration: allowing slightly higher spectral width or peak amplitude (subject to hardware constraints) can offset decoherence effects, particularly dephasing. Conversely, if spectral width is the limiting factor, shortening the gate duration or reweighting the relevant penalty in the cost functional may prevent the observed saturation and recover fidelity without excessive spectral spreading. When the above is taken into account, high-fidelity gate operations can be achieved in the experiments.

We will now discuss the results of our optimized quantum control simulations in the context of the experimentally available parameter range. We focus on the most studied platform of Ti atoms deposited on ultrathin layers of MgO grown on a silver substrate. Currently, Ti atoms on 2 monolayers (ML) and 3 ML of MgO are available according to experimental results. The spin lifetime of Ti has been studied on 2 ML of MgO and lies around $T_1 = 150$ ns (for the remote qubits); for 3 ML, Ti has been studied by ESR-STM⁴⁹, but the lifetime has not been established. Based on the difference in lifetimes for Fe on 2 and 3 ML of MgO however¹⁸, we can expect at least one order of magnitude increase in T_1 . We therefore set $T_1 \leq 1000$ ns as a conservative estimate for 3 ML of MgO. The most commonly reported parameters are summarized in Table 2.

Here, we parametrized the drift Hamiltonian using the Larmor frequencies and upper/lower bounds of coupling strengths from ref. 1. The

coupling strength between remote qubits is chosen so that resonant frequencies are well separated. To explore the limits of this platform, we set the temperature $T = 0.1$ K, which is achievable in a dilution fridge, and consider energy relaxation as the relevant decoherence mechanism. We assume that Ti atoms on 3 ML can achieve at least $T_1^{(2)} = 1000$ ns and set the coherence time of the sensor qubit to $T_1^{(1)} = 10$ ns. To account for realistic driving strengths, the pulse width is set to $\tau = 12.5$ ns, which is the shortest possible pulse width (strongest driving) realized in experiments. We consider two scenarios for the quantum-coherent manipulation. First, we used a *DC pulsing scheme* where the DC bias of the sensor spin is only active after the quantum control during the readout phase (see Fig. 6b). In this case the sensor and remote spins are subject to the same relaxation mechanism governed by the spin-substrate energy relaxation and we set $T_1^{(1)} = T_1^{(2)} = T_1^{(3)} = 1000$ ns, which corresponds to the lowest energy relaxation rates as shown in Fig. 3b. In such a case, one can expect a high gate fidelity as the coherence time is much longer than the pulse width. This corresponds to the protocol suggested in previous theoretical works⁵⁰, but is not the one used in recent experiments. Instead, refs. 1,8,10 used a *DC always-on* scheme where the DC bias is on during quantum gate operations and readout phase (mostly due to technical reasons related to drift compensation), as illustrated in Fig. 6c. In such a scenario, we investigate the realization of the CNOT gate on two remote qubits as illustrated in Fig. 6a. Using the Krotov method, the CNOT gate fidelity is improved from 0.446 to 0.887 when using the DC pulsing scheme. The result suggests that the DC pulsing scheme is highly beneficial to achieve high-fidelity quantum gates. Notice that the spectrum of the optimized pulse here also shows four pronounced peaks at the resonant frequencies, similar to the closed-system case (see Supplementary Material Fig. S2).

The distribution of coherent and diagonal fidelities over the Liouvillian basis provides a further insight into the performance of quantum gates, compared to a mere average gate fidelity. Therefore, we further analyze the components of the average gate fidelity. The average gate fidelity can be

written in the following form⁵¹:

$$F_{\text{avg}} = \frac{1}{N(N+1)}(F_{\text{coh}} + F_{\text{dia}}),$$

$$F_{\text{coh}} = \sum_{i,j=1}^N \langle \phi_i | U_{\text{target}}^\dagger \rho_{ij} U_{\text{target}} | \phi_j \rangle,$$

$$F_{\text{dia}} = \sum_{i,j=1}^N \langle \phi_i | U_{\text{target}}^\dagger \rho_{jj} U_{\text{target}} | \phi_i \rangle.$$
(13)

where $|\phi_i\rangle$ denotes the i -th *basis state*, and $\rho_{ij} = \mathcal{E}(|\phi_i\rangle\langle\phi_j|)$ is the output obtained by applying the dynamical map $\mathcal{E}$ to the Liouvillian basis operator $|\phi_i\rangle\langle\phi_j|$ (we refer to this one as the *forwarded state*). $U_{\text{target}}^\dagger$ is the desired unitary transformation, and N denotes the dimension of the Hilbert space of the drift Hamiltonian. It is worth breaking down the average fidelity into two parts, as shown in Eq. (13): $\mathcal{F}_{\text{coh}}$ quantifies the preservation of coherence, while $\mathcal{F}_{\text{dia}}$ quantifies the population fidelity. Coherent fidelity can take complex values, yet only the real part is relevant, as the imaginary part will vanish when summing over all basis states.

As shown in Fig. 6c, although the average fidelity is 0.446, the fidelities associated with individual state transitions exhibit noticeable variation. This observation indicates that optimizing each transition separately within the state-to-state approach can yield higher fidelities compared to the gate optimization.

With the DC pulsing scheme and thus long coherence times for all qubits, both coherent and diagonal fidelities are high and nearly uniform across the Liouvillian basis. However, without the DC pulsing scheme and thus short coherence time for the sensor qubit, the coherent and diagonal fidelities are significantly reduced and skewed towards low-energy states $|000\rangle$ and $|\overline{001}\rangle$. This is due to energy relaxation, where higher-energy states decay into lower-energy ones. With a short coherence time of sensor qubit $T_1^{(1)} = 10$ ns, the Krotov method can only perform well for low-energy states, while the fidelity for high-energy states is significantly reduced. In addition, the distribution of diagonal fidelity is asymmetric. This feature is a result of the unidirectional energy relaxation.

In this work, we investigated how quantum optimal control theory (QOCT), implemented through Krotov-based optimization, interacts with the dynamical constraints of static exchange-coupled spin qubits. Closed-system analyses serve as useful reference points, where QOCT reaches gate

Table 2 | Experimentally reported lifetime T_1 times of Ti sensor and remote spins on MgO/Ag(100). ML stands for monolayers

Substrate	T_1^{Sensor} (ns)	T_1^{Remote} (ns)	T (K)	Refs.
2 ML	$5 < T_1 < 10$	$100 < T_1 < 150$	0.4	1,8,10
3 ML	$5 < T_1 < 10$	$T_1 \geq 1000$	0.4	18,49

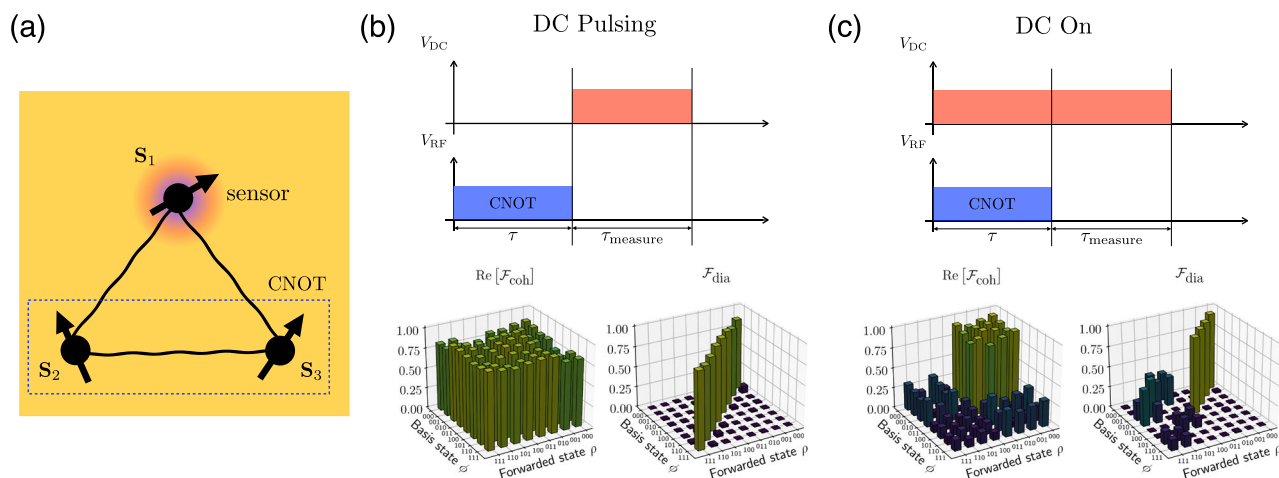


Fig. 6 | Performance of QOCT with realistic parameters. **a** CNOT operation is implemented on two remote qubits. **b** Coherent and diagonal fidelity with using DC pulsing scheme, such that lifetimes are $T_1^{(1)} = T_1^{(2)} = T_1^{(3)} = 1000$ ns. The average gate fidelity is $\mathcal{F}_{\text{avg}} = 0.887$. **c** Coherent and diagonal fidelity without using DC pulsing scheme, $T_1^{(1)} = 10$ ns and $T_1^{(2)} = T_1^{(3)} = 1000$ ns. The average gate fidelity is

$\mathcal{F}_{\text{avg}} = 0.446$. The parameters of the drift Hamiltonian are $\omega_1 = 17$ GHz, $\omega_2 = 15$ GHz, $\omega_3 = 13$ GHz, $J_{12} = 0.1$ GHz, $J_{23} = 0.35$ GHz, $J_{13} = 0.6$ GHz. Temperature is $T = 0.1$ K, and the pulse width is $\tau = 12.5$ ns. See the definition of basis state and forwarded state in the main text below Eq. (13).

fidelities close to $\mathcal{F} = 100\%$, almost independent of the drift Hamiltonian parameters. The more relevant regime for surface-based qubits is one in which asymmetrical relaxation and dephasing are explicitly incorporated into the control landscape. When these open-system effects are included, the optimized control fields exhibit characteristic spectral changes that depend on the dominant decoherence mechanism. We observe consistent improvement of fidelity and concurrence with QOCT compared to unoptimized or Rabi-synchronized driving. For coherence parameters that are realistic for current experiments (remote-qubit lifetimes exceeding 1 μs and weak pure dephasing), fidelities above 90% appear feasible using experimentally accessible pulse shapes and DC bias pulsing. While extending noise-aware QOCT to larger systems increases computational cost and pulse complexity, the essential advantage persists when the optimization incorporates the decoherence channels that dominate the platform rather than relying on closed-system targets. In this light, the fidelity gains obtained under experimentally motivated pulsing schemes should be viewed as indications of how control strategies benefit from aligning pulse design with open-system dynamics. Future investigations into scalability and robustness against parameter fluctuations may clarify how these methods extend beyond the present setting, and where the balance between model completeness and practical control performance lies for future implementations.

Finally, we remark that based on our results, improvements in the gate performance of this platform can be achieved along the following trajectories: First, enhancing the T_1 time of the remote qubits (Fig. 3b), which can be achieved by better decoupling between the qubits and the metallic substrate⁴⁹. Second, increasing the driving strength thereby reducing the gate time τ with respect to the lifetime will reduce the influence of noise during the gate operation. This can be achieved experimentally by optimizing the local driving field⁵². Third, lowering the temperature to maintain a high state polarization for a given splitting (Fig. 4c). Combined with the DC pulsing scheme presented here, gate-fidelities exceeding $\mathcal{F} \gtrsim 0.9$ should be achievable straightforwardly.

Methods

Numerical simulations

All numerical simulations were carried out using the QuTip⁵³ package (Version 4.7.6) together with the Krotov⁵⁴ Python package (Version 1.3.0). These packages provide efficient tools for quantum dynamics simulations and gradient-based optimal control.

Computation of time-evolution operators of closed systems

We solve the time evolution using *Trotterization* for closed quantum systems. Suppose the duration for the control pulse is τ , we divide the total time into N_{seg} segments, each with duration $\Delta t = \tau/N_{\text{seg}}$. The time-evolution operator for each segment is approximated as

$$U(t_{n+1}) = \exp[-iH(t_n)\Delta t]U(t_n), \quad (14)$$

where $U(0) = I$ is the initial unitary operator and $H(t_n)$ is the Hamiltonian at time $t_n \equiv n\Delta t$. The full time-evolution operator is then given by the product of all segments

$$U(\tau) = \prod_{n=0}^{N_{\text{seg}}-1} \exp[-iH(t_n)\Delta t]. \quad (15)$$

The approximation becomes exact in the limit of $N_{\text{seg}} \rightarrow \infty$. In our simulations, we choose N_{seg} such that $\Delta t = 0.01$ ns and the results converge.

Computation of spin dynamics of closed systems

Suppose the initial density matrix of the two-qubit system is $\rho(0)$, the density matrix at time t is given by

$$\rho(t) = U(t)\rho(0)U^\dagger(t), \quad (16)$$

with $U(t)$ being the time-evolution operator. The spin of the second qubit at time t is computed as

$$\langle S_2^\alpha(t) \rangle = \text{Tr}_1[S^\alpha \rho(t)], \quad (17)$$

where $\alpha = x, y, z$ and Tr_1 is the partial trace over the first qubit. In the interaction picture, one needs to perform an additional transformation to the density matrix as

$$\rho_I(t) = e^{iH_0 t} \rho(t) e^{-iH_0 t}, \quad (18)$$

where H_0 is the drift Hamiltonian.

Computation of gate fidelity of closed systems

In the lab frame, the evolution that corresponds to the drift Hamiltonian should be taken into account. As for the NOT gate on the second qubit, the target unitary operator is given by⁵⁵

$$U_{\text{target}}(t) = \exp(-iE_1 t)|00\rangle\langle 01| + \exp(-iE_2 t)|01\rangle\langle 00| + \exp(-iE_3 t)|10\rangle\langle 11| + \exp(-iE_4 t)|11\rangle\langle 10|. \quad (19)$$

The fidelity of the gate operation is then computed using

$$\mathcal{F}_{U(t), U_{\text{target}}} = \frac{|\text{Tr}[U(t)^\dagger U_{\text{target}}(t)]|^2 + d}{d(d+1)}, \quad (20)$$

where d is the dimension of the Hilbert space (for example, $d = 4$ for a two-qubit system).

Quantum optimal control theory with the Krotov method

Quantum optimal control theory (QOCT) is a set of methods to devise external control fields that drive the system toward a desired objective. Assuming the system is described by the following time-dependent Hamiltonian

$$\hat{H}(t) = \hat{H}_0 + \sum_j u_j(t)\hat{H}_j, \quad (21)$$

where $\hat{H}_0$ is the time-independent drift Hamiltonian and $u_j(t)$ are time-dependent control functions. The goal is to solve the set of control fields, defined over a duration τ , to achieve a specific purpose, such as implementing a desired unitary operation (i.e., quantum gate) or transition from an initial to a target state. The former is called *gate optimization*, and the latter is *state-to-state optimization*. The gate optimization is essential for realizing universal quantum gates, where the operation must be correct for any of the basis states, while the state-to-state optimization targets a specific transition from an initial to a target state over the control time. We particularly employ the Krotov method for our work since it is well-suited for both closed and open quantum systems, and is efficiently implemented in the Python Krotov library⁵⁴.

A minimal introduction to the Krotov method for our two-qubit system in Eq. (2) with a single control field can be presented as follows. The Krotov method can be formulated as an optimization process to minimize the following cost functional form

$$J[|\phi_k^i(t)\rangle, u^i(t)] = J_T(|\phi_k^i(\tau)\rangle) + \int_0^\tau g_a(u^i(t))dt, \quad (22)$$

where $|\phi_k^i(t)\rangle$ are the time-evolved initial states $|\phi_k^i(0)\rangle$ under the control field $u^i(t)$ at iteration i . (Indeed, there exists an additional term to penalize state population in a subspace, yet we don't employ it in our work.) For state-to-state optimization, $k = 1$, while for gate optimization, $k = 1, 2, 3, 4$ for the four basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. The first term is the main functional cost, which essentially measures the distance between the desired unitary

transformation and actual evolution. Given a set of initial states $|\phi_k\rangle$ to evolve to a set of target states $|\phi_k^{\text{target}}\rangle$, their distance can be quantified by the fidelity

$$f_k = \langle \phi_k^{\text{target}} | \phi_k(\tau) \rangle, \quad (23)$$

and the function J_T reads

$$J_T = 1 - \frac{1}{N} \text{Re} \left[\sum_{k=1}^N f_k \right]. \quad (24)$$

The second term is the running cost functional, which penalizes the deviation of the control field from a reference field $u_{\text{ref}}(t)$, and is defined as

$$\begin{aligned} g_a(u^i(t)) &= \frac{\lambda_a}{S(t)} [u^i(t) - u_{\text{ref}}^i(t)]^2; \\ u_{\text{ref}}^i(t) &= u^{i-1}(t), \end{aligned} \quad (25)$$

with λ_a being the inverse step width and $S(t)$ is an “update shape” function that enforces smooth on/off switches. In our simulations, we choose $u_{\text{ref}}^0(t)$ as a flatpulse with smooth turn-on and turn-off edges, and $S(t)$ is also a flatpulse⁵⁴. The choice of λ_a is determined at the beginning of the optimization process and is kept constant throughout the iterations.

The above description works for the closed quantum system, i.e., without decoherence. In the presence of decoherence, we adopt the method in ref. 45 to incorporate the Lindblad master equation and the three representative states of Liouville space. In this case, the cost functional is modified as

$$J_T = 1 - \sum_{i=1}^3 \frac{w_i}{\text{Tr}[\hat{\rho}(0)]} \text{Re} \left\{ \text{Tr} \left[\hat{U}_{\text{target}} \hat{\rho}_i(0) \hat{U}_{\text{target}}^\dagger \hat{\rho}_i(T) \right] \right\}, \quad (26)$$

where $\hat{\rho}_i$ ($i = 1, 2, 3$) denote three encoded density matrices of an orthonormal basis of Liouville space, and w_i are the corresponding weights satisfying normalization condition. The choice of w_i is also determined at the beginning of the optimization process, and is kept constant throughout the iterations. The dimension of Liouville space is 2^{2N} for N qubits.

Optimization parameters

The control optimization was performed using the Krotov method, which ensures monotonic convergence of the cost functional. The time evolution was discretized into N_t time steps with $\Delta t = \tau/N_t$, where τ is the total propagation time. Throughout this work, we set $\Delta t = 0.01$ ns, which is approximately two-times smaller than the smallest time scale in the system (i.e., the inverse of the largest Larmor frequency of the Hamiltonian), to ensure numerical convergence. Influence of undersampling/oversampling time segment Δt on the optimized fidelity is discussed in the Supplementary Material Fig. S11. Propagation was performed using the exponential matrix propagator `expm` for closed-system dynamics and the built-in ordinary differential equations solver `DensityMatrixODEPropagator` for open-system calculations, both implemented within the Krotov package. A flatpulse update shape function $S(t)$ was employed to enforce smooth boundary conditions on the control fields. The step-size parameter λ_a was chosen separately for each optimization to balance the update strength with numerical stability. In particular, we used $\lambda_a = 10$ for NOT gate optimization and $\lambda_a = 0.01$ for Bell-state preparation.

Convergence criteria

The convergence can be monitored through changes in the optimization functional J_T , which depends on the fidelity as defined in Eqs. (23) and (24) and corresponds to the infidelity. In closed-system optimizations, unit fidelity is known to be attainable, and we define convergence as $J_T \leq 10^{-6}$. Alternatively, convergence can be conditioned on the change in J_T , $\Delta J_T = |J_T^N - J_T^{N-1}| \leq 10^{-6}$ between successive iterations, which means that fidelity

became stationary. For open-system dynamics, the maximum achievable fidelity is lower than 1, and it is hard to predict a reliable target fidelity. Instead, we found it more stable to run the optimization for a fixed number of iterations, $N_{\text{iter}}^{\text{max}} = 600$ for closed systems and $N_{\text{iter}}^{\text{max}} = 100$ for open systems, and verified that the infidelity remained stationary ($\Delta J_T \leq 10^{-6}$) over a significant number of iterations, which was the case in all simulations. This way, we ensured a balance between computational time and optimization outcome.

Concurrence

Given a two-qubit density matrix ρ , the concurrence is defined as⁵⁶

$$C(\rho) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}, \quad (27)$$

where the λ_i are the eigenvalues, arranged in decreasing order, of the following Hermitian matrix

$$R = \sqrt{\sqrt{\rho} \tilde{\rho} \sqrt{\rho}}, \quad (28)$$

where

$$\tilde{\rho} \equiv (\sigma'_y \otimes \sigma'_y) \rho^* (\sigma'_y \otimes \sigma'_y), \quad (29)$$

with ρ^* the complex conjugate of ρ . Here σ'_y denotes the Pauli- y matrix in eigenstate basis, with

$$\sigma'_y \otimes \sigma'_y = -|00\rangle\langle 11| + |\tilde{0}\tilde{1}\rangle\langle \tilde{1}\tilde{0}| + |\tilde{1}\tilde{0}\rangle\langle \tilde{0}\tilde{1}| - |11\rangle\langle 00|. \quad (30)$$

Collapse operators

We used collapse operators acting on the eigenstates of the system as defined in ref. 10. Description for collapse operators can be found in the Supplementary Material.

Data availability

The datasets generated during the current study are not publicly available because they require analysis scripts that are not yet curated for public release, but are available from the corresponding author on reasonable request.

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Author contributions

C.W. and H.-A.L. conceived the ideas of the manuscript; H.-A.L., S.T., and D.J. performed the numerical simulations. All authors contributed to writing the manuscript under the supervision of C.W.

Competing interests

The authors declare no competing interests.

Additional information

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